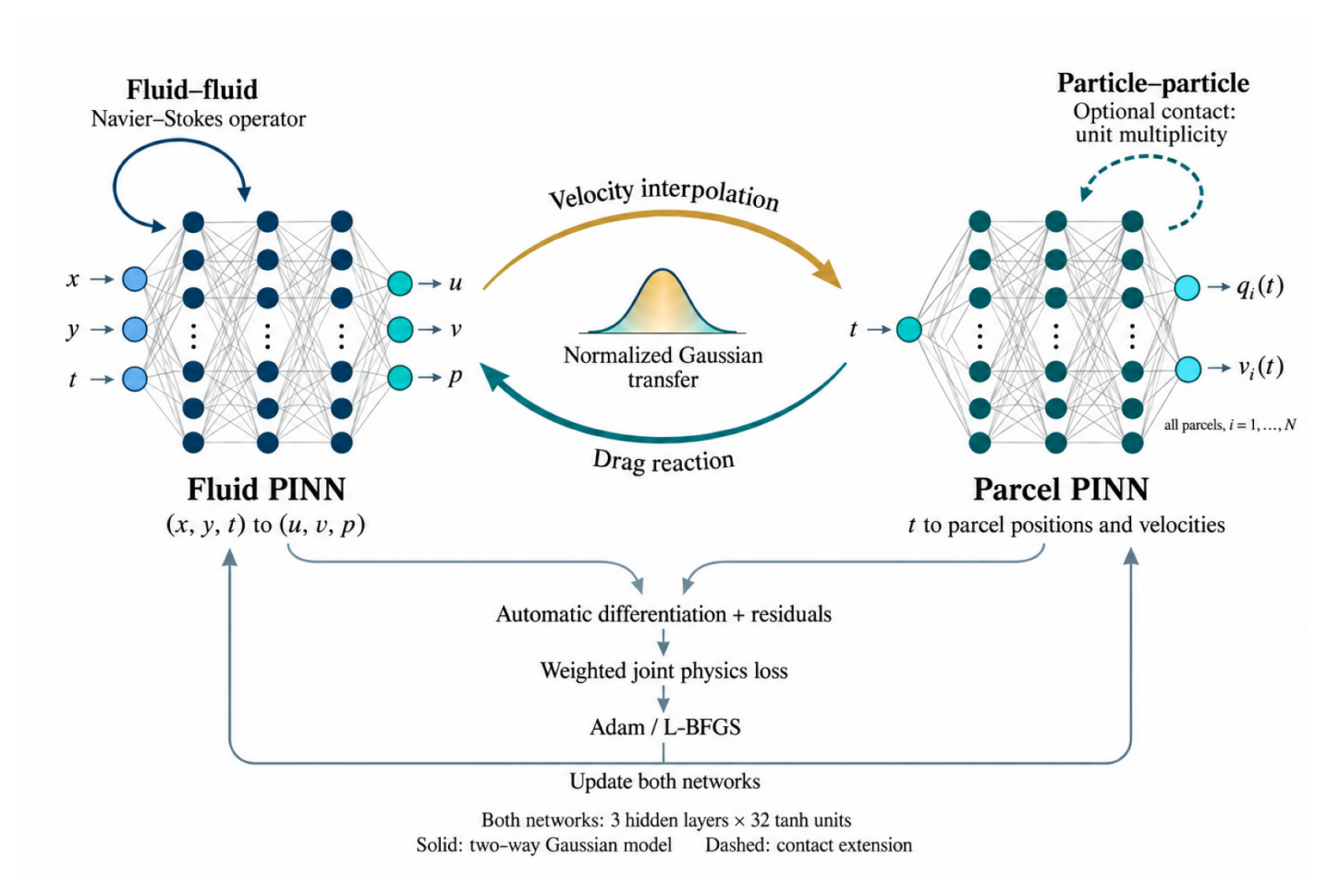

Fluid–fluid
Navier–Stokes operator
Particle–particle
Optional contact:
unit multiplicity
Velocity interpolation
Normalized Gaussian
transfer
Drag reaction
x
y
t
u
v
p
t
$q_i(t)$
$v_i(t)$
all parcels, $i = 1, \ldots, N$
Fluid PINN
(x, y, t) to (u, v, p)
Parcel PINN
t to parcel positions and velocities
Automatic differentiation + residuals
Weighted joint physics loss
Adam / L-BFGS
Update both networks
Both networks: 3 hidden layers × 32 tanh units
Solid: two-way Gaussian model
Dashed: contact extension